scientific reports

Check for updates

OPEN

Machine learning-based forecasting of ground surface settlement induced by metro shield tunneling construction

Qiankun Wang[1], Chuxiong Shen[1]✉, Chao Tang[2], Zeng Guo[1], Fangqi Wu[1] & Wenyi Yang[1]

Urban rail transit systems, represented by subways, have significantly alleviated the traffic pressure brought by urbanization and have addressed issues such as traffic congestion. However, as a commonly used construction method for subway tunnels, shield tunneling inevitably disturbs the surrounding soil, leading to uneven ground surface settlement, which can impact the safety of nearby buildings. Therefore, it is crucial to promptly obtain and predict the ground surface settlement induced by shield tunneling construction to enable safety warnings and evaluations. This study collects multi-point surface settlement data from monitoring sections and proposes a data preprocessing method based on tangent circles to transform discrete monitored data into continuous and smooth data. On this basis, the Particle Swarm Optimization (PSO) algorithm is employed to optimize a Back Propagation Neural Network(BPNN) for the subsequent prediction of ground surface settlement. The influence of network structure and prediction mode on prediction accuracy is investigated. The study predicts and verifies the settlement over a 5-day period during both the slow settlement stage and the stable stage to evaluate the predictive performance of the proposed method. The maximum relative error of the prediction based on the data preprocessing method and PSO-BP algorithm is 0.46%, demonstrating a good fitting effect. The study indicates that this method can provide a valuable reference for safety warnings and control of ground surface settlement during shield tunneling construction.

Keywords Shield construction, Metro tunneling, Machine learning, Ground settlement

With the acceleration of urbanization, a series of 'urban diseases", such as traffic congestion, caused by population aggregation have emerged. A well-developed subway transportation network has, to some extent, alleviated traffic congestion and improved travel convenience for residents. Over the past few decades, subway projects in densely populated cities have developed rapidly. However, subway construction can have potentially disastrous impacts on nearby buildings, structures, and public facilities, making it essential to carefully choose construction methods to maximize the safety of surrounding infrastructure[1]. Additionally, subway tunnels often pass beneath existing lines, rivers, lakes, and important buildings or structures, where complex underground hydrogeological conditions and surrounding environmental factors make safety accidents more likely. The shield tunneling method, known for its fast excavation speed, high degree of automation, and minimal impact on the ground surface, has been widely applied in subway tunnel construction projects[2].

With the acceleration of urbanization, a series of 'urban diseases", such as traffic congestion, caused by population aggregation have emerged. A well-developed subway transportation network has, to some extent, alleviated traffic congestion and improved travel convenience for residents. Over the past few decades, subway projects in densely populated cities have developed rapidly. However, subway construction can have potentially disastrous impacts on nearby buildings, structures, and public facilities, making it essential to carefully choose construction methods to maximize the safety of surrounding infrastructure1. Additionally, subway tunnels often pass beneath existing lines, rivers, lakes, and important buildings or structures, where complex underground hydrogeological conditions and surrounding environmental factors make safety accidents more likely. The shield tunneling method, known for its fast excavation speed, high degree of automation, and minimal impact on the ground surface, has been widely applied in subway tunnel construction projects2.

[1]School of Civil Engineering and Architecture, Wuhan University of Technology, Wuhan 430070, China. [2]Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572025, China. ✉email: 242684@whut.edu.cn

However, the concealed nature of subway shield tunneling projects and the complexity of the surrounding environment mean that safety incidents during subway shield construction occur from time to time. As a complex and high-risk project, safety issues in shield tunneling still require significant attention[3]. The safety issues in subway shield tunneling construction can be broadly categorized into three types: worker safety, tunnel structural safety, and surrounding environmental safety[4]. Worker safety refers to the threats faced by construction personnel during operations, such as falling from heights, being struck by objects, mechanical injuries, and electric shocks. Tunnel structural safety involves issues like segment flotation, face instability, and structural deformation during shield tunneling construction. Surrounding environmental safety pertains to the impact of tunnel construction on the nearby environment, leading to a series of safety problems, primarily manifesting as ground settlement. The first type of safety issue is mainly controlled by enhancing workers' awareness of safety practices. The second and third types of safety issues are primarily addressed through comprehensive analysis based on site monitoring. Shield tunneling inevitably disturbs the surrounding soil, especially in areas with soft clay. The disturbed soil is subjected to compression, shearing, and twisting, leading to elastoplastic deformation or even failure, eventually resulting in ground surface settlement. Soil settlement can cause uneven settlement of existing foundations of surrounding buildings, leading to safety accidents[5,6]. During shield tunneling, as the construction progresses, the surrounding soil is disturbed, leading to a redistribution of soil stress and causing ground settlement. This disturbance forms a "settlement trough", affecting urban roads, buildings, and pipelines within the settlement area, resulting in differential settlement. When differential settlement exceeds the threshold, it may lead to damage, causing personal injury and property loss. Current research on predicting ground settlement induced by shield tunneling focuses on the final settlement, while there is insufficient analysis of the settlement process, which is crucial for controlling settlement during construction. However, conducting trend analysis on the ground settlement induced by shield tunneling and predicting changes in the settlement trough can aid in the safety management of the construction process, enabling dynamic early warnings and improving safety management efficiency. Therefore, it is necessary to promptly acquire data on ground surface settlement caused by shield tunneling construction and predict the settlement process of the soil, providing a reference for safety warnings.

This paper achieves continuous processing of monitoring data based on data preprocessing methods and applies machine learning algorithms to predict ground settlement induced by metro shield tunneling construction. Compared with previous studies, this paper focuses on the settlement process rather than the total settlement prediction. The main contribution of this study lies in the establishment of a continuous time-series dataset for ground settlement, combining the PSO and BPNN algorithms to realize dynamic predictions of ground settlement during shield tunneling construction. The remainder of this paper is organized as follows. Section 2 summarizes related work. Section 3 describes the relevant methods, including the Back Propagation Neural Network(BPNN), the Particle Swarm Optimization-based BP neural network (PSO-BP), and data preprocessing methods. Section 4 presents the project overview and details the data collection and preprocessing. The application results of the relevant methods are discussed in Sect. 5. Finally, Sect. 6 concludes the main findings of this study.

Literature review

Traditional method of settlement prediction

The factors influencing ground settlement mainly include tunnel parameters, geotechnical properties, and construction effects. There is a complex nonlinear relationship between ground settlement and these factors. Tunnel parameters include the depth, shape, and excavation area of the tunnel. Geotechnical properties refer to the physical characteristics of the soil, which are the decisive factors for ground settlement. Construction effects refer to human factors during the tunneling process. For example, in shield tunneling, ground settlement can be controlled by adjusting face pressure, excavation speed, or the amount of grouting[7]. Accurately predicting ground surface settlement is of great significance for construction safety. Existing studies have recognized the importance of settlement prediction, with current research mainly utilizing empirical formulas, numerical simulations, and model tests for prediction. Empirical formulas are approximate expressions obtained by fitting field-measured data. In terms of empirical formulas, ground settlement caused by shield tunneling can be predicted based on several mechanical theories, including elastic half-space theory, random medium theory, the Perk formula, the limit equilibrium method, and the energy conservation theory. Zhang combined the improved elastic layered half-space model with a method that considers the inhomogeneity of the soil. Using the displacement control boundary integral method, Zhang studied the effect of soil stratification on ground surface settlement caused by tunnel excavation. A comparative analysis was conducted using the finite element method to verify the effectiveness of the proposed approach[8]. Yang optimized a simplified ground movement prediction method based on random medium theory. A comparative analysis of the results obtained from the optimized method and the original theoretical method showed that the optimized method could effectively predict ground movement[9]. Divall, considering the impact of bidirectional tunnel stress superposition, proposed a modified overlapping zone based on empirical formulas to correct soil settlement caused by bidirectional tunnels[10]. Although empirical formulas provide a convenient way to predict ground settlement during shield tunneling construction, the factors considered in theoretical research are limited, leading to certain deviations between the calculated results and the actual outcomes. Therefore, the application of empirical formula methods in practical engineering is limited. Numerical simulation methods discretize the strata for analysis, allowing for consideration of the interaction between the shield machine and the soil, as well as complex factors such as construction parameters. These methods are widely used for predicting settlement during shield tunneling construction. The main numerical simulation methods include the finite element method (FEM), the finite difference method (FDM), and three-dimensional nonlinear methods. Xie used the three-dimensional finite difference method to simulate the shield machine tunneling process. Using the Yingbin Third Road Tunnel in

Shanghai as an example, the effectiveness of this numerical model and its applicability in large-diameter tunnels were validated through field measurements[11]. Liu established a three-dimensional model of key nodes in the structure of a subway station using FLAC(3D) software, analyzing the mechanical performance of these key nodes and their impact on ground settlement. The feasibility of the method and the accuracy of the numerical simulation were verified by comparing with field monitored data[12]. Shivaei studied the interaction mechanism between mechanized twin tunnel construction and groundwater using three-dimensional coupled finite element analysis (FEA). Field data on ground settlement were used to calibrate the baseline numerical model[13]. Li, based on biomechanics and rheology theories, established a three-dimensional fluid-solid full coupling mathematical model of soil deformation induced by shield tunneling. A comparative analysis with measured settlement data from Chengdu Metro Line 4 was conducted to verify the correctness of the mathematical model[14]. However, numerical simulation has difficulties in precisely determining the parameters of the surrounding soil, the construction constitutive model, and boundary conditions. The constitutive model often differs significantly from the actual soil, and the results of numerical simulations are easily affected by mesh division, leading to generally moderate prediction accuracy. Model tests can account for a wide range of parameters and effectively replicate the interactions during shield tunneling construction. Ritter proposed an experimental simulation method that uses powder-based three-dimensional (3D) printing to create building models. This method quantifies the model effects and boundary conditions, successfully reproducing the soil-structure interactions during tunnel excavation[15]. However, Huang argues that although model tests can consider more parameters, the various parameters involved in shield tunnel construction are difficult to replicate, making it impossible to achieve identical construction conditions[16]. Therefore, it is evident that model tests cannot fully replicate the settlement patterns of the soil, resulting in limited prediction accuracy. Additionally, the cost of model tests is a significant limiting factor.

Machine learning-based method of settlement prediction

Ground surface settlement caused by subway shield tunneling construction is a complex, multi-factor, nonlinear problem. Traditional methods such as empirical formulas, numerical simulations, and model tests have significant limitations, making it challenging to achieve accurate settlement predictions. Moreover, traditional methods mainly focus on the prediction of the total settlement of stabilized soil, with insufficient attention given to the process of settlement during shield tunneling construction. Machine learning is a method of implementing intelligent data analysis using computers, and it has strong high-dimensional nonlinear fitting capabilities, offering advantages over traditional methods in predictive tasks. As one of the most popular artificial intelligence methods, machine learning can utilize a large amount of historical data for training to establish relationships between given information (input) and expected outcomes (output), without the need to assign physical meanings to parameters. The main machine learning prediction algorithms include Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM) networks, Deep Belief Networks (DBN), and others. Existing research has already recognized the importance of predicting ground surface settlement, and many scholars have conducted related studies based on machine learning. Chen, using monitored data from the Changsha Metro Line 4 project as a database, applied three types of artificial neural networks to predict the maximum ground settlement of an EPB shield tunnel. The results indicated that the Generalized Regression Neural Network (GRNN) showed significant advantages in accuracy and computation time compared to the Back Propagation Neural Network(BPNN) and the Radial Basis Function (RBF) Neural Network[17]. Xu established a building settlement prediction model for subway construction using an LSTM network, studying the influence of construction parameters, engineering geological parameters, and spatial parameters on the prediction accuracy of building settlement[18]. Xiao used Fiber Bragg Grating (FBG) sensors to monitor soil settlement caused by shield tunneling construction. By introducing a wavelet-based signal denoising method to preprocess the monitored data for smoother results, a Time Series-Based Back Propagation Neural Network (TS-BPNN) was used for subsequent soil settlement prediction. The study explored the impact of model structure and database size on prediction accuracy[19]. Zhang proposed a simulation-hybrid method that combines artificial intelligence, simulation, and experimentation. The simulation component aims to study the complex tunnel-soil interactions, using Support Vector Machines (SVM) to predict ground surface settlement caused by tunnel excavation[20]. Mahmoodzadeh conducted a study based on 1,800 settlement data samples from eight urban shield tunnels in Iran, comparing seven intelligent prediction methods: Long Short-Term Memory (LSTM), Deep Neural Networks (DNN), k-Nearest Neighbors (KNN), Gaussian Process Regression (GPR), Support Vector Regression (SVR), Decision Trees (DT), and Linear Regression (LR)[21]. Zhang collected 7,800 settlement data records from shield tunneling construction projects and used this data to evaluate the predictive performance of algorithms such as Extreme Gradient Boosting (XGBoost), Multivariate Adaptive Regression Splines (MARS), Artificial Neural Networks (ANN), and Support Vector Machines (SVM)[22]. On this basis, the performance of these four algorithms was further compared using 1100 monitored data points from three shield tunneling construction projects in Singapore. The results indicated that the prediction accuracy of XGBoost was slightly higher than that of the other three algorithms[23]. Moghtader considered ten parameters related to ground settlement and used Back Propagation Artificial Neural Network (BP-ANN) technology to predict the ground surface settlement for Tehran Metro Line 6, determining the relative impact of different parameters on tunnel-induced ground surface settlement[24]. Ocak employed Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Gaussian Processes (GP) to predict ground surface settlement caused by shield tunneling construction. In combination with a real engineering case, the prediction error rates for the three methods were 13%, 12.8%, and 9%, respectively[25]. Elbaz proposed a model based on Deep Reinforcement Learning (DRL), which serves as a valuable tool to improve the performance of the shield machine's drive system (i.e., thrust and cutterhead torque). Elbaz also introduced a novel optimal model based on reinforcement learning, integrating the Q-learning network with the metaheuristic Grey Wolf Optimizer to enhance the accuracy of shield tunneling path prediction. Elbaz proposed a model based on Deep Reinforcement

Learning (DRL), which serves as a valuable tool to improve the performance of the shield machine's drive system (i.e., thrust and cutterhead torque)[26]. Bouayad proposed a method combining Principal Component Analysis (PCA) with an Adaptive Neuro-Fuzzy Inference System (ANFIS) to model the nonlinear relationship between ground settlement induced by Earth Pressure Balance Tunnel Boring Machines and geological parameters[27].

However, traditional machine learning algorithms are often constrained by the temporal and spatial complexity of feature learning, leading to significant prediction errors. Some scholars have sought to improve prediction accuracy by optimizing the initial parameters of the predictive algorithms. Ge posited that ground surface settlement is influenced by three main factors: geological conditions, tunnel geometry, and shield operation factors, and established an index system comprising 12 indicators to reflect these influences. Based on 370,404 engineering data points from the construction of the Guangzhou Metro Line 18 shield tunnel, Ge proposed a deep learning method using a Deep Belief Network (DBN) optimized by the Whale Optimization Algorithm (WOA) to predict ground surface settlement[28]. Su suggested that ground surface settlement is affected by 11 key factors. Using Bayesian Optimization to fine-tune the hyperparameters of the XGBoost algorithm, Su analyzed 533 ground surface settlement monitored data points from a shield tunneling construction project in a city and compared the optimized algorithm with Artificial Neural Networks (ANN) and Support Vector Machines (SVM). The results indicated that the XGBoost algorithm, optimized by Bayesian methods, achieved higher prediction accuracy[7]. Zhang combined Wavelet Packet Transform (WPT) with the Least Squares Support Vector Machine (LSSVM) and validated the feasibility and applicability of the WPT-LSSVM method using actual monitored data from the Wuhan Metro. The results showed that the tunnel daily settlement predictions based on the WPT-LSSVM method were more accurate and reliable[29]. Moghaddasi proposed a hybrid model based on an Artificial Neural Network (ANN) optimized by the Imperialist Competitive Algorithm (ICA). The accuracy of the proposed ICA-ANN model was validated using actual monitored data from Line 2 of the Karaj Metro in Iran[30]. Zhang categorized the main variables affecting ground settlement into two broad categories: geological condition parameters and shield operation parameters, and used a combination of Differential Evolution Algorithm(DEA) and Artificial Neural Network(ANN) to predict ground settlement[31]. Zhang conducted a comprehensive comparison of the predictive performance of Back Propagation Neural Network(BPNN), Generalized Regression Neural Networks (GRNN), Extreme Learning Machines (ELM), Support Vector Machines (SVM), and Random Forests (RF). The global optimal hyperparameters of the machine learning algorithms were determined using Particle Swarm Optimization(PSO), thereby enhancing the settlement prediction performance of the algorithms[32]. Tang optimized the hyperparameters of Support Vector Machines (SVM), Random Forests (RF), Backpropagation Neural Networks (BPNN), and Deep Neural Networks (DNN) using Particle Swarm Optimization (PSO) and Grid Search (GS). The results indicated that the optimized models could accurately and effectively predict settlement, with the RF model demonstrating superior prediction accuracy across different datasets compared to the other three methods[33]. Lin optimized a Long Short-Term Memory (LSTM) neural network using the Particle Swarm Optimization (PSO) algorithm. The model was tested and trained using actual settlement data collected from a shield tunnel in Shenzhen. The results showed that the optimized algorithm achieved the best performance, providing guidance for processing measured data from Earth Pressure Balance (EPB) shield machines[34]. Yazdanparast combined image processing algorithms with Artificial Neural Network (ANN) technology to predict the ground surface settlement of Qom Metro Line A[35]. Elbaz introduced a novel optimal model based on reinforcement learning, integrating the Q-learning network with the metaheuristic Grey Wolf Optimizer to enhance the accuracy of shield tunneling path prediction[36]. Liu integrated Finite Element Analysis (FEA) with machine learning algorithms, introducing intelligent optimization algorithms to invert geological parameters and synchronous grouting variables, thereby predicting ground settlement caused by synchronous grouting. The effectiveness of the proposed method was validated using actual settlement data from a shield tunnel in Beijing[37].

In combination with previous studies, it can be observed that the prediction of ground surface settlement induced by subway tunneling shield construction primarily focuses on the final settlement after stabilization, analyzing the relationship between the total settlement and related parameters, with relatively little research on process settlement during construction. Additionally, due to the complexity of construction projects and the discrete nature of monitoring, the real-time monitored data obtained are often sparse and small in quantity, making it challenging to achieve accurate and effective settlement predictions. To predict the process settlement of soil during subway tunnel shield construction and improve the accuracy of predictions, this paper proposes a data preprocessing method based on tangent circles. The Particle Swarm Optimization (PSO) algorithm is used to optimize the structure of the Back Propagation Neural Network(BPNN). Based on the preprocessed data, machine learning algorithms are applied to predict ground surface settlement.

Method

Back propagation neural network—BPNN

The Back Propagation Neural Network (BPNN) is a type of supervised learning neural network and is currently one of the most widely used artificial neural networks[38]. In a BP neural network, the processing units are neurons, and a typical neuron model generally consists of three parts: connection weights and thresholds, a summation unit, and an activation function. The connection weights are used to store learned knowledge, while the threshold determines the excitation state of the neuron. The learning ability of a BP neural network model is achieved by continuously adjusting the connection weights and thresholds. The summation unit performs the weighted summation of input signals. The activation function processes the information transformation and mainly includes linear functions, ramp functions, and sigmoid functions. The basic structure of a BP neural network consists of an input layer, a hidden layer, and an output layer, with each layer composed of many simple neurons that operate in parallel. The number of hidden layers is variable, and the layers are connected by

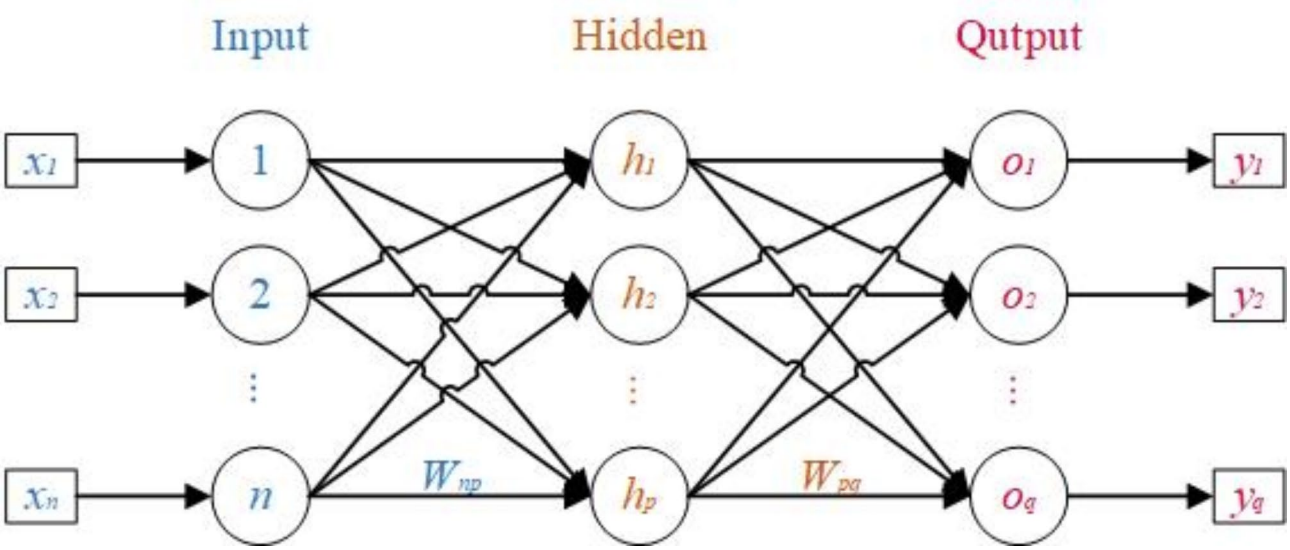


Fig. 1. The structure of BPNN.

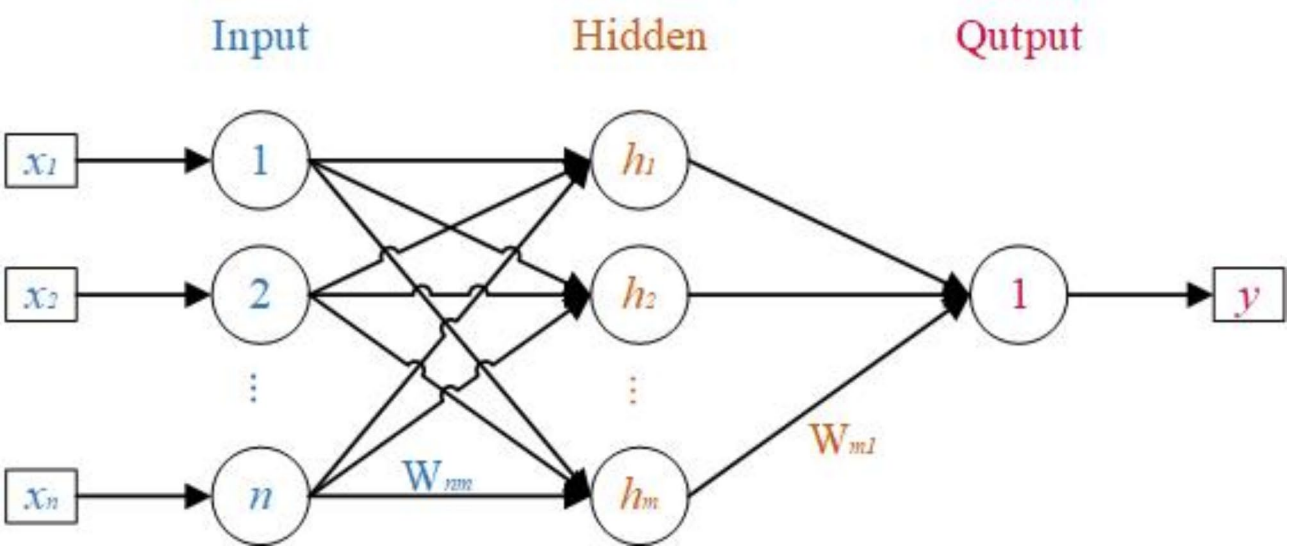


Fig. 2. The structure of BPNN.

learnable weights. Neurons in different layers do not have direct connections and are only connected to adjacent neurons. A typical structure of a BP neural network is shown in Fig. 1.

The BP neural network divides the training process into two stages: forward propagation of signals and backward propagation of errors. During the forward propagation of signals, external signals are transmitted from the input layer through the hidden layers to the output layer. The state of each layer's neurons only affects the neurons in the next layer, and the output signal is generated at the output end. During this process, the network's weights remain fixed. If the output layer does not produce the desired output, the process enters the error backward propagation stage. In the error backward propagation process, the error signal starts from the output end and propagates backward layer by layer, adjusting the connection weights of each neuron through error feedback. Through repeated forward and backward propagation, the weights between neurons are continuously adjusted and corrected. By alternating these two processes, the gradient descent strategy of the error function is executed in the weight vector space to minimize the network's error function. The algorithm terminates when the actual output at the output end falls within an acceptable range of the desired output.

Time series-based BP neural network—TS-BPNN

The structure of the Time Series-Based BP Neural Network (TS-BPNN) is represented as n-m-1. Here, *n* denotes the number of input layer nodes, *m* represents the number of hidden layer nodes, and in this study, there is only one output layer node, as shown in Fig. 2.

In the TS-BPNN, the first n values are used as inputs, and the $(n+1)$th value is used as the output. The values of n and m need to be determined through testing. In the TS-BPNN, the range of hidden layer nodes generally follows the principle:

$$m = \sqrt{n+1} + \alpha \tag{1}$$

where α is a constant between 0 and 10.

New settlement prediction values can be generated through forecasting, based on which two soil settlement prediction modes can be established, as shown in Fig. 3. In the figure, P_n represents the settlement value at monitoring point T_n. The light green indicates the monitored values, while the light blue indicates the predicted values. Each row in the figure contains $(n+1)$ values, with the first n values used as inputs and the $(n+1)$th value as the output. Mode 1 relies entirely on the first n monitoring values to predict the subsequent $(n+1)$th settlement value, with new predictions being made only when new monitored data is obtained. Mode 2 involves iterative prediction, where the predicted $(n+1)$th value is used as a new input to predict the $(n+2)$th settlement value based on the 2nd to $(n+1)$th values. Mode 1 depends entirely on the existing monitored data, generally yielding higher prediction accuracy, but with a limited number of prediction steps; in this study, the prediction step count is limited to one. Mode 2 allows for long-term prediction by using the new predicted value as input, but this also results in cumulative errors, making the prediction accuracy generally lower than that of Mode 1[19].

Optimizing the BP neural network using the PSO algorithm—PSO-BP

The BP neural network algorithm lacks global search capability, as it is a single-point search method based on the gradient descent of the error function. During the learning and training process of the BP neural network, it has certain drawbacks, such as poor robustness, slow convergence speed, and weak generalization ability. The advantage of the Particle Swarm Optimization (PSO) algorithm lies in its ability to search for the global optimum based on the currently discovered optimal solution. The PSO algorithm has a simple structure, is easy to implement, requires minimal parameter tuning, and has a fast convergence speed with a wide search range. It is currently widely used in neural network training and other fields[39].

In the process of BP neural network modeling, the learning ability of the BP neural network model is achieved by continuously adjusting the connection weights and thresholds. Therefore, the weights w and thresholds θ have a significant impact on the training results; they are a set of random values and can easily fall into local minima[40]. Training the weights and thresholds is essentially a complex problem of finding the optimal parameters. Using the gradient descent method to update the connection weights and thresholds can lead to completely different results in the neural network even with small changes. By optimizing the BP neural network using the PSO algorithm, it is necessary to encode the weights and thresholds between neurons in the BP neural network structure. Intelligent search is then conducted for each particle to find the most suitable weights and thresholds, enabling the BP neural network to achieve faster convergence and better generalization ability.

The PSO algorithm optimized BP neural network model is shown in Fig. 4 and includes the following steps:

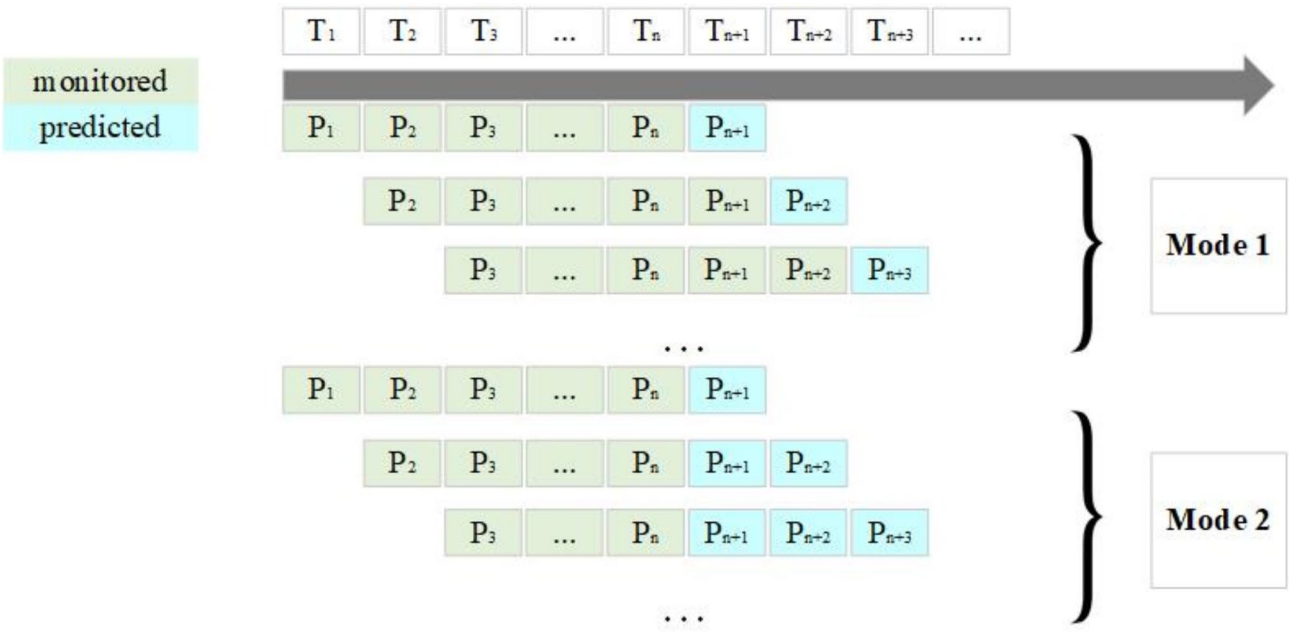


Fig. 3. Comparison of data prediction modes.

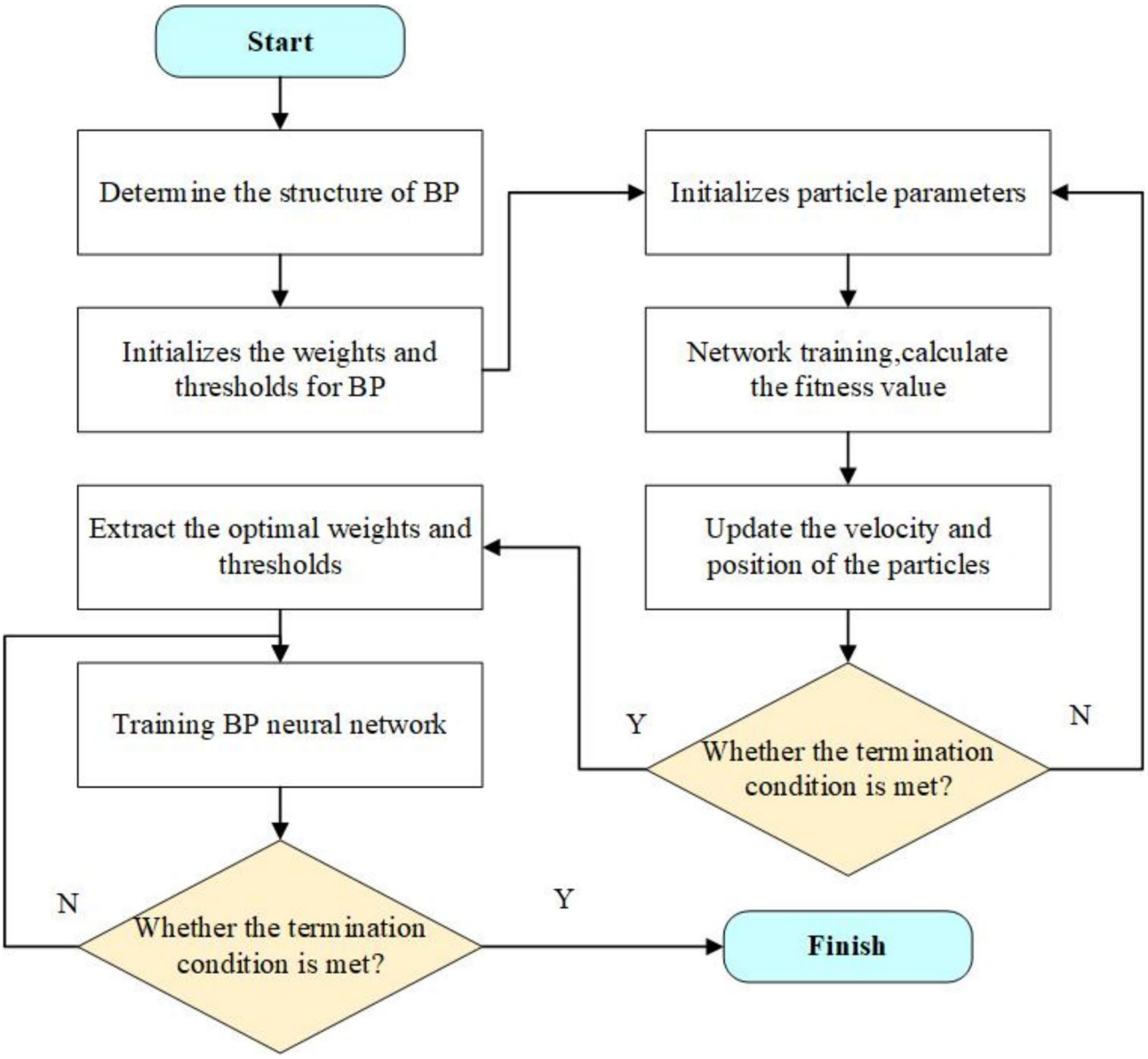


Fig. 4. PSO-BP neural network algorithm flowchart.

1. Determine the structure of the BP neural network: Initialize the weights and thresholds of the BP neural network.
2. Initialize particle parameters: This includes particle velocity, particle position, individual best value, global best value, inertia weight, and learning factors.
3. Calculate fitness value: Train the neural network model, and use the error value obtained from training as the fitness value of the particles.
4. Find the individual historical best and global best values and update parameters: By comparing the fitness value of each particle with the previous best position and the global best position, continuously update the velocity and position of the particles.
5. Obtain the global best value: After reaching the termination condition, obtain the optimal weights and thresholds for the objective function. If the termination condition is not met, continue training until the global optimal solution is found.
6. BP neural network training: After obtaining the weights and thresholds optimized by PSO, use the BP neural network for training until the termination condition is satisfied.

Data preprocessing method

In the actual process of subway shield tunneling construction, the monitoring frequency for ground surface settlement by the construction party or third-party monitors is generally once a day, and the monitoring period for a specific point usually does not exceed three months. As a result, the amount of data collected is relatively small. However, predicting ground surface settlement using machine learning methods often requires a large

amount of monitored data. This is also why there are more studies on predicting the total settlement after stabilization rather than the process settlement during construction. Nevertheless, in construction management, predicting process settlement can sometimes be more important than predicting total settlement. Therefore, it is necessary to adopt data preprocessing methods to expand the discrete monitored data and transform them into a continuous and smooth data stream. Many studies have been conducted on data augmentation methods. In subway construction, commonly applied methods include interpolation and wavelet transform. Based on relevant literature at home and abroad and the characteristics of shield tunnel construction, this study provides a tangent circle-based data preprocessing method for ground surface settlement, which converts discrete monitored data into continuous and smooth monitored data. As shown in Fig. 5, for a monitoring point, the settlement values at four monitoring time points T_1,T_2,T_3 and T_4 are V_1,V_2,V_3andV_4, respectively. The slope of the line segment V_1-V_2 is X_{1-2}, the slope of the line segment V_2-V_3 is X_{2-3}, and the slope of the line segment V_3-V_4 is X_{3-4}. The angle between line segments V_1-V_2 and V_2-V_3 is Angle 2, and the angle between line segments V_2-V_3 and V_3-V_4 is Angle 3. Bisectors of Angles 2 and 3 intersect at point O_{2-3}. Using this point as the center, a circle is drawn, and the arc passing through points S_2 and S_3 represents the settlement process data at the monitoring point during time T_2-T_3.

Collection of monitored data

Project overview

This study analyzes the settlement induced by shield tunneling construction on Wuhan Metro Line 12. Wuhan Metro Line 12 is a loop line that originates and ends at Wuchang Railway Station, crossing the Yangtze River twice. It is the first loop metro line in Wuhan, with a total length of approximately 59,876 m (entirely underground) and 37 stations. This project focuses on monitoring ground surface settlement caused by the excavation of the shield tunneling construction in the LJ to LW section. The shield tunnel includes two lines, the Left Line and the Right Line, with an outer diameter of 6.2 m and an inner diameter of 5.5 m. At the topmost layer is the filling material, with uneven thickness; followed by layers of clayey silt and mud clay, each approximately 2 m to 2.5 m thick. Below is a thicker layer of mucky silty clay, around 12 m, which surrounds the tunnel above; beneath the

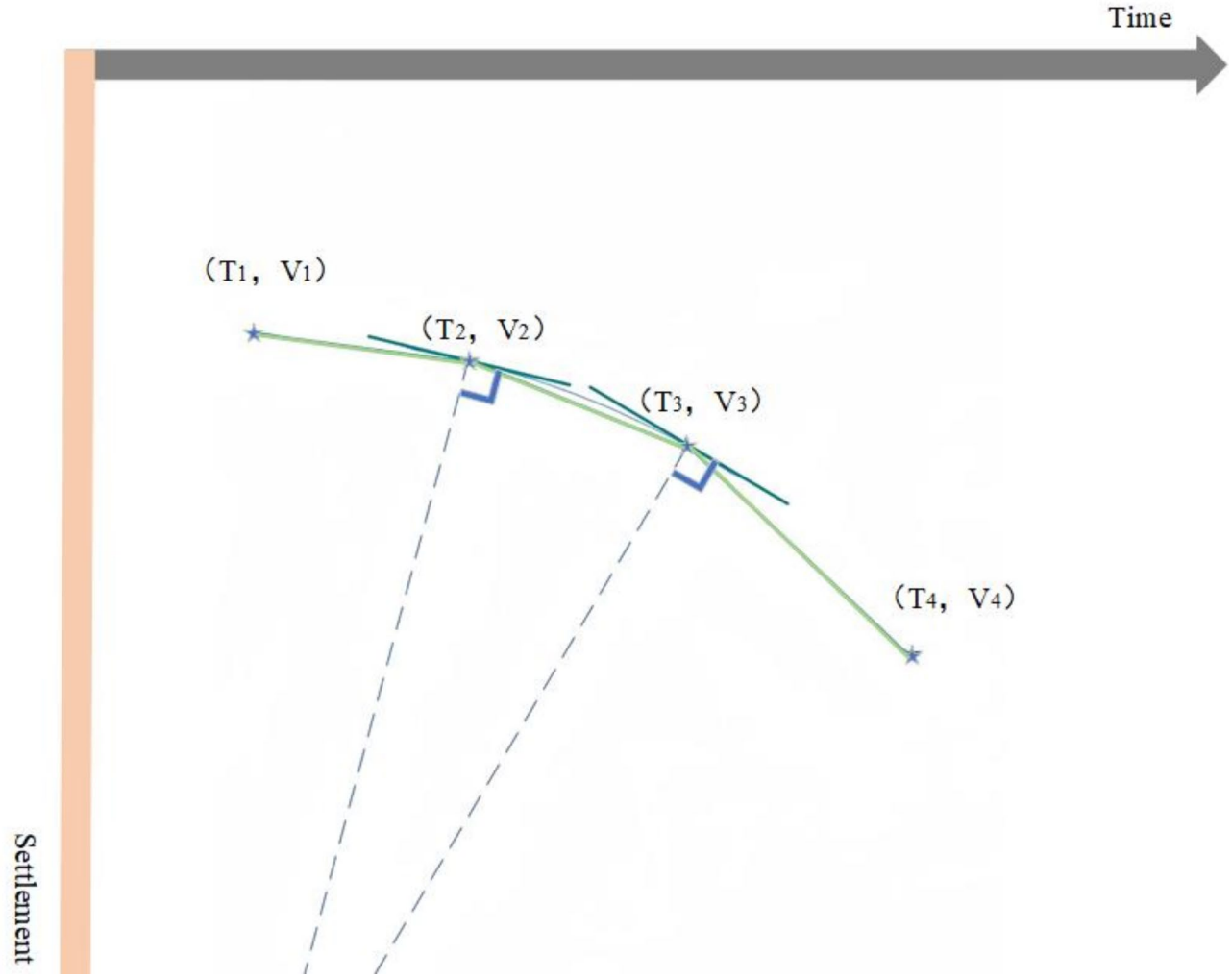


Fig. 5. Schematic diagram of tangent circle-based data preprocessing method.

tunnel lies silty clay. The center-to-center distance between the left and right lines is 12.6 m, and the depth of the overburden layer is 12.3 m. A specific section of the LJ to LW tunnel, where complete settlement monitored data is available and has been verified by third-party monitoring, was selected as the study subject. In this section, five monitoring points, DB-1, DB-2, DB-3, DB-4, and DB-5, were set up as shown in Fig. 6. Additionally, numerous monitoring points were established around the tunnel to ensure real-time monitoring of the overall safety of the tunnel.

Preprocessing of the monitored data

Monitoring was conducted at the five monitoring points from June 1, 2024, to August 9, 2024, for a total duration of 70 days. The scatter plots of ground surface settlement at each monitoring point are shown in Fig. 7. It can be observed that the ground surface settlement process can be generally divided into four stages: the uplift stage, the rapid settlement stage, the slow settlement stage, and the stable stage. On June 1, the shield machine was about to reach the monitoring section, causing a phenomenon of soil uplift, which continued until June 6, when the ground surface had mostly returned to its initial state. From June 7, rapid settlement began, with a maximum settlement rate of up to 2.95 mm/day. Starting from June 16, the settlement rate decreased to below 0.5 mm/day, entering the slow settlement stage. From July 4, the settlement rate further decreased to below 0.1 mm/day, reaching a generally stable state.

To increase the data volume, data from the latter half of the uplift stage, the rapid settlement stage, the slow settlement stage, and the stable stage were preprocessed. Using the tangent circle-based data preprocessing method mentioned in Sect. 3.4, the discrete monitored data were transformed into continuous and smooth monitored data. The processed data are shown in Fig. 8.

Analysis of the monitored data

To gain a more intuitive understanding of the settlement changes at the monitoring section during different stages, the settlement troughs of several representative days based on the monitored data are presented in Fig. 9. It can be observed that during the uplift stage, the ground surface exhibits significant fluctuations and an irregular shape. In the rapid settlement stage, as time progresses, it can be seen that each monitoring point is undergoing a process of rapid settlement. In the slow settlement and stable stages, the settlement monitored data exhibit a 'V' shape, where the further a monitoring point is from the central axis of the shield tunnel, the lower the settlement value.

Results and analysis

Analysis of the effect model structure

Due to the lack of interpretability of the Artificial Neural Network (ANN) method, repeated experiments are needed to determine the optimal model structure. To identify the best model structure, the ground surface settlement data from monitoring point DB-3 were used as a sample since DB-3 is located directly above the tunnel crown, making its values representative of the settlement profile. Based on the original data and the preprocessed data, and in accordance with relevant machine learning literature[19], the data volume was expanded threefold. Considering the constraint on the range of hidden layers described in Eq. (1), the number of nodes in the input and hidden layers were set to 5, 10, and 15 for cross-validation. Thus, nine BP neural network models and nine PSO-BP neural network models were selected for comparison. The hyperparameter settings of the model are shown in Table 1.

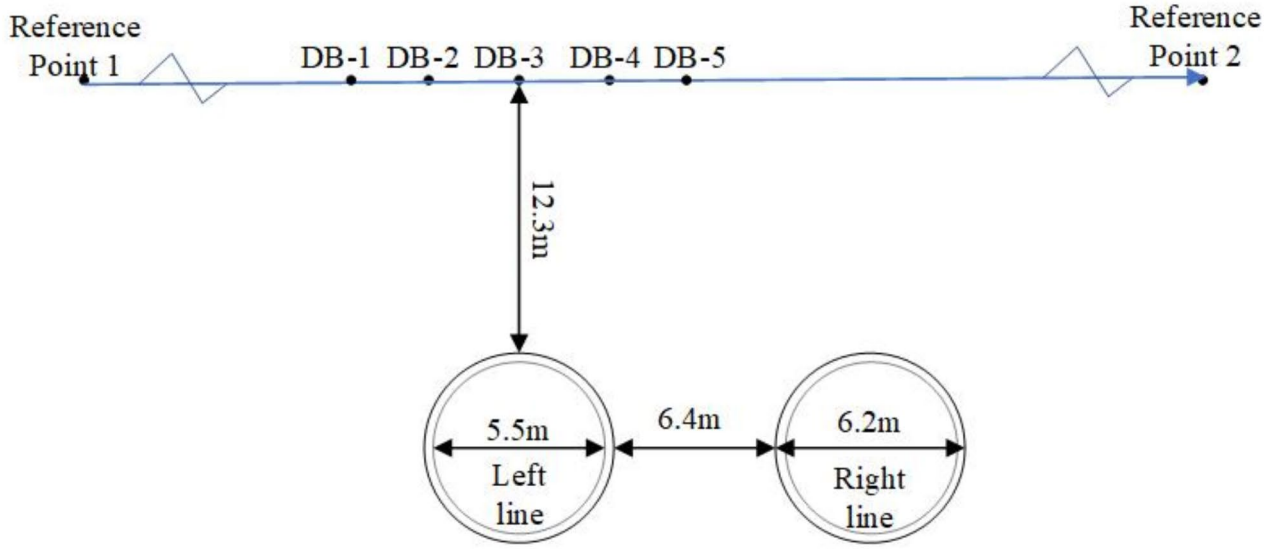


Fig. 6. Basic information of the project.

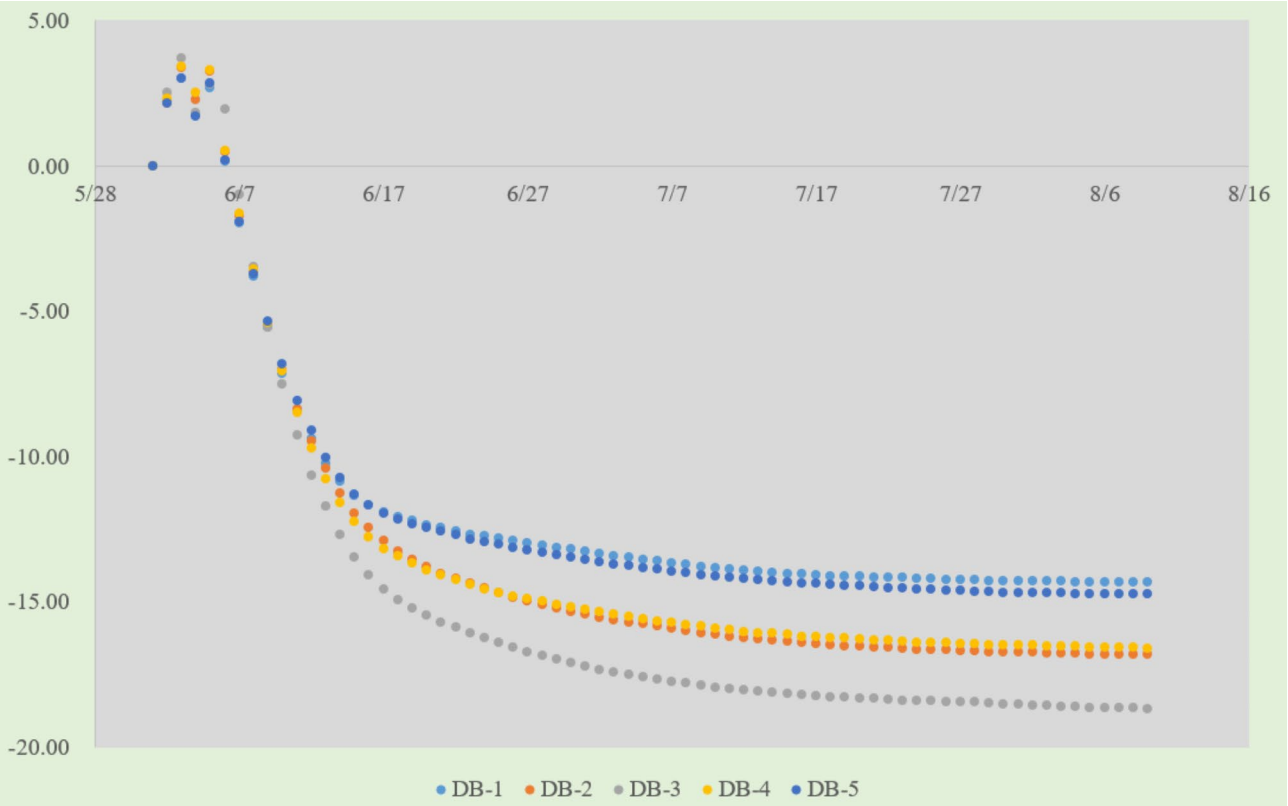


Fig. 7. Scatter plots of ground surface settlement at monitoring points.

In terms of evaluation indexs, the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) between the predicted and monitored values were used as evaluation standards to assess the network's performance. The formulas for calculating RMSE and MAE are shown in Eqs. (2) and (3).

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(P_i - M_i)^2} \tag{2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|P_i - M_i| \tag{3}$$

The comparative study results of the BP neural network prediction experiments are shown in Fig. 10. The results indicate that when the input layer, hidden layer, and output layer have 10, 10, and 1 node(s), respectively, the BP neural network achieves the best prediction performance, with RMSE and MAE of 0.0366 and 0.0250, respectively.

The comparative study results of the PSO-BP neural network prediction experiments are shown in Fig. 11. The results reveal that when the input layer, hidden layer, and output layer have 10, 10, and 1 node(s), respectively, the PSO-BP neural network achieves the smallest RMSE. When the input layer, hidden layer, and output layer have 15, 5, and 1 node(s), respectively, the PSO-BP neural network achieves the smallest MAE.

To select the best PSO-BP neural network structure, this study calculated the error coefficient *RS* using a standardization method. The formulas for calculating *RS* is shown in Eq. (3).

$$RS_i = \frac{RMSE_i - RMSE_{\min}}{RMSE_{\max} - RMSE_{\min}} + \frac{MAE_i - MAE_{\min}}{MAE_{\max} - MAE_{\min}} \tag{4}$$

Finally, the error coefficients for each network structure were obtained, as shown in Table 2.

It can be observed that when the input layer, hidden layer, and output layer have 10, 10, and 1 node(s), respectively, the PSO-BP neural network achieves the best prediction performance, with RMSE and MAE of 0.0116 and 0.0213, respectively. Compared to the BP neural network, the PSO-BP neural network reduced RMSE and MAE by 68.19% and 14.89%, respectively. The results indicate that the PSO-BP neural network significantly improves prediction accuracy.

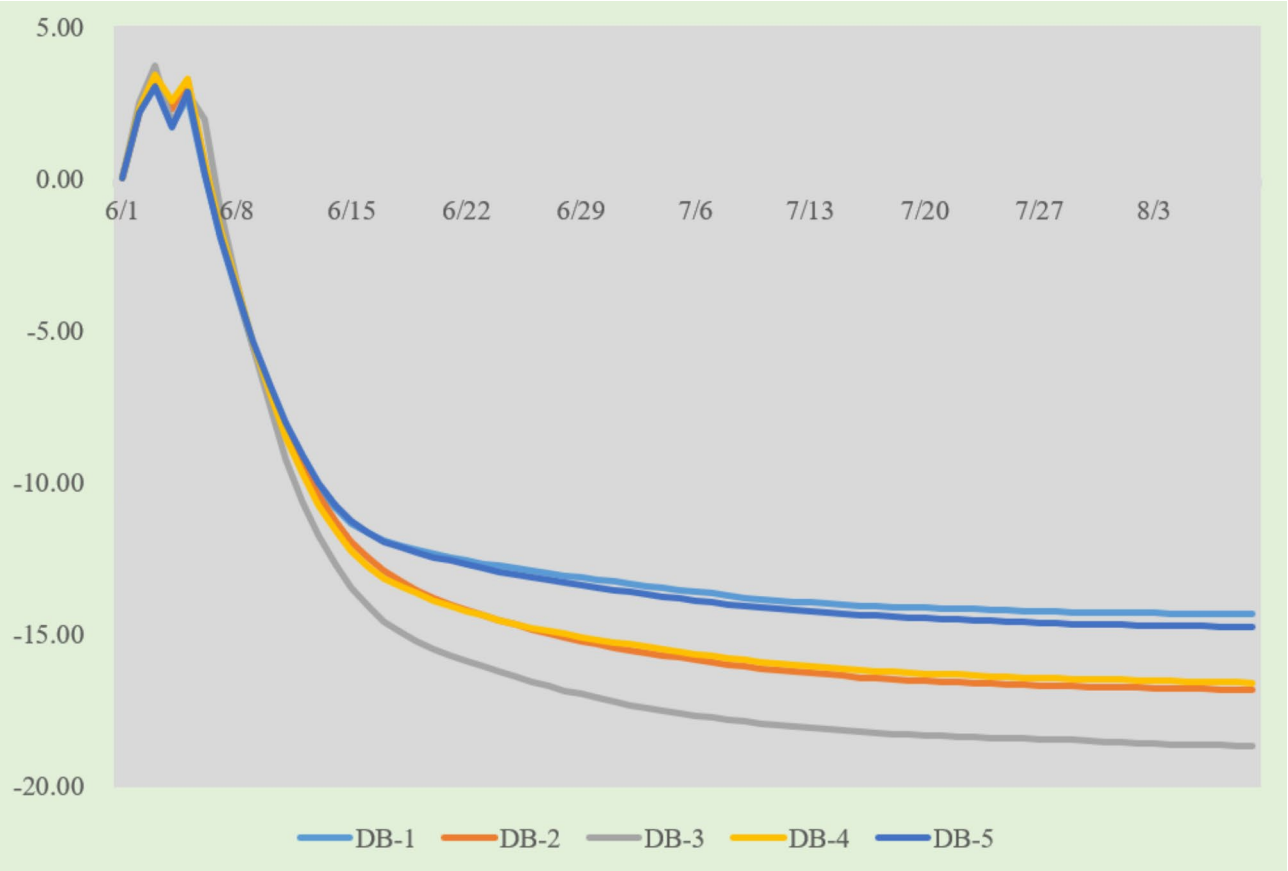


Fig. 8. Preprocessed settlement data at monitoring points.

Additionally, predictions of ground surface settlement at DB-3 were made using the original data. Using the 10-10-1 network structure, prediction analysis was performed with both the original BP neural network and the PSO-BP neural network. The relevant results are shown in Table 3. It can be observed that when predicting with the original data, both the original BP neural network and the PSO-BP neural network exhibit significant errors, further demonstrating the effectiveness of the data preprocessing method proposed in this paper for data prediction.

Comparison of the prediction modes

As previously mentioned, there are two prediction modes. Mode 1 relies entirely on the first n monitored values to predict the subsequent $(n+1)$th settlement value, with new predictions being made only when new monitoring data is obtained. Mode 2 uses the newly predicted settlement values as input data, allowing for longer-term predictions of future settlement. Based on the data from DB-3, a prediction step of 10 was set. Using the original BP and PSO-BP algorithms, a comparative study of prediction accuracy was conducted for both modes. The prediction trends are shown in Fig. 12, and the evaluation results are presented in Fig. 13.

It can be observed that, for both the original BP neural network and the PSO-BP algorithm model, the predictions based on complete monitoring data (Mode 1) are superior to those based on partially new predicted data (Mode 2). However, despite the lower prediction accuracy of Mode 2 compared to Mode 1, the error remains within an acceptable range, allowing for an acceptable prediction. Moreover, since Mode 2 provides long-term settlement predictions, further prediction research is conducted using Mode 2.

Verification of the settlement prediction

Based on the PSO-BP algorithm and Mode 2, settlement predictions were implemented for early warning purposes, and these predictions were validated using monitoring results. Additionally, we conducted a comparison with the Autoregressive Integrated Moving Average (ARIMA) model proposed by Box et al. Since the time periods for the uplift and rapid settlement stages are relatively short, predictions were made for the slow settlement stage and the stable stage.

For the slow settlement stage, the training data for the PSO-BP algorithm consisted of the preprocessed data from the beginning of the settlement stage up to the prediction period. The ARIMA1 algorithm used the same data as the PSO-BP algorithm, while the ARIMA2 algorithm used the latter half of the data. For the stable stage, the training data for the PSO-BP algorithm consisted of the preprocessed data from the beginning of the settlement stage up to the prediction period. Considering the longer duration of the stable stage, the dataset for

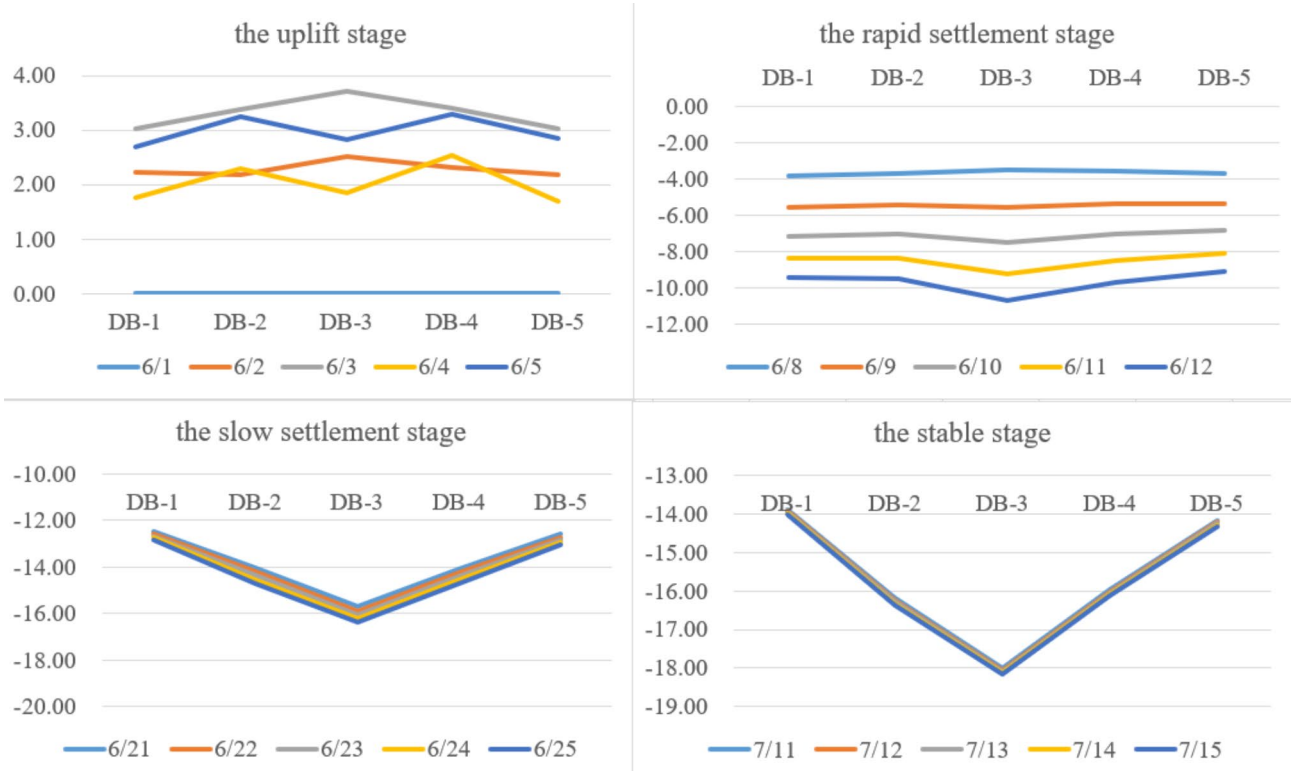


Fig. 9. Representative Settlement Troughs of DB-1 to DB-5.

Hyperparameter	Value
Learning rate	0.00001
Max_iteration	20
Training epochs	300

Table 1. Model hyperparameter settings.

the ARIMA algorithms was further split. The ARIMA3 algorithm used the same data as the PSO-BP algorithm, ARIMA4 used the latter half of the data, and ARIMA5 used the last quarter of the data.

The prediction results for DB-1 to DB-5 during the slow settlement stage are shown in Fig. 14. It can be observed that the prediction performance of the ARIMA algorithm during the slow settlement stage is significantly lower than that of the PSO-BP algorithm, with ARIMA1 performing worse than ARIMA2.

Statistical analysis was conducted using the Mean Relative Error (MRE) and the Largest Relative Error (LRE). The formulas for calculating MRE and LRE are shown in Eqs. (5) and (6). The results are presented in Tables 4 and 5.

$$MRE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{P_i - M_i}{M_i}\right| \tag{5}$$

$$LRE = \max_{i=1}^{n}\left|\frac{P_i - M_i}{M_i}\right| \tag{6}$$

Statistical analysis was conducted using the Mean Relative Error (MRE) and the Largest Relative Error (LRE). The formulas for calculating MRE and LRE are shown in Eqs. (5) and (6). The results are presented in Tables 4 and 5. The MRE ranges for PSO-BP, ARIMA1, and ARIMA2 are 0.13%~0.27%, 3.56%~12.11%, and 0.49%~6.00%, respectively. The LRE for PSO-BP, ARIMA1, and ARIMA2 are 0.46%, 20.05%, and 13.41%, respectively. The PSO-BP prediction results exhibit a smaller range of relative errors, with the lowest LRE among the three methods, and the ARIMA2 method outperforms ARIMA1.

The prediction results for DB-1 to DB-5 during the stable stage are shown in Fig. 15. It can be observed that, at this stage, the PSO-BP algorithm and the ARIMA algorithm each have their strengths and weaknesses in predicting ground surface settlement. The MRE and LRE for each algorithm are shown in Tables 6 and 7. The MRE ranges for PSO-BP, ARIMA3, ARIMA4, and ARIMA5 are 0.04%~0.10%, 3.24%~5.82%, 0.10%~0.66%, and

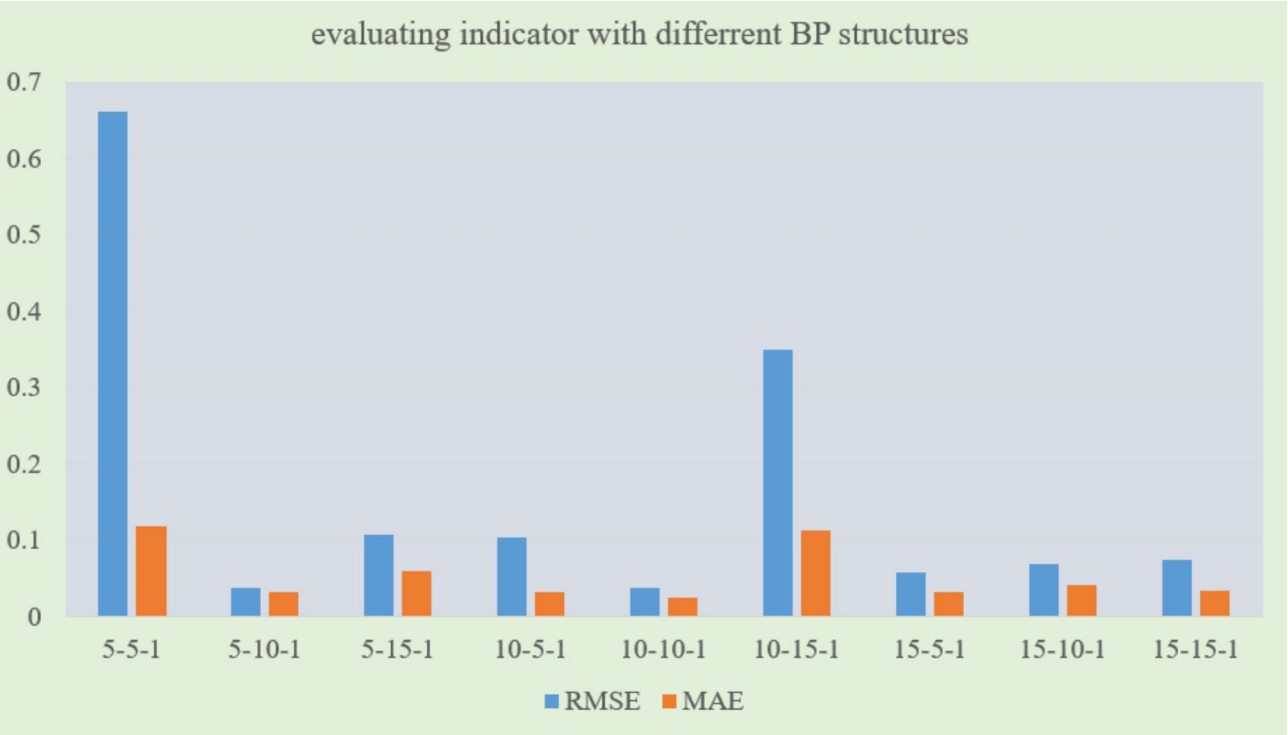


Fig. 10. Comparison of the settlement prediction of DB-3 with different BP network structures.

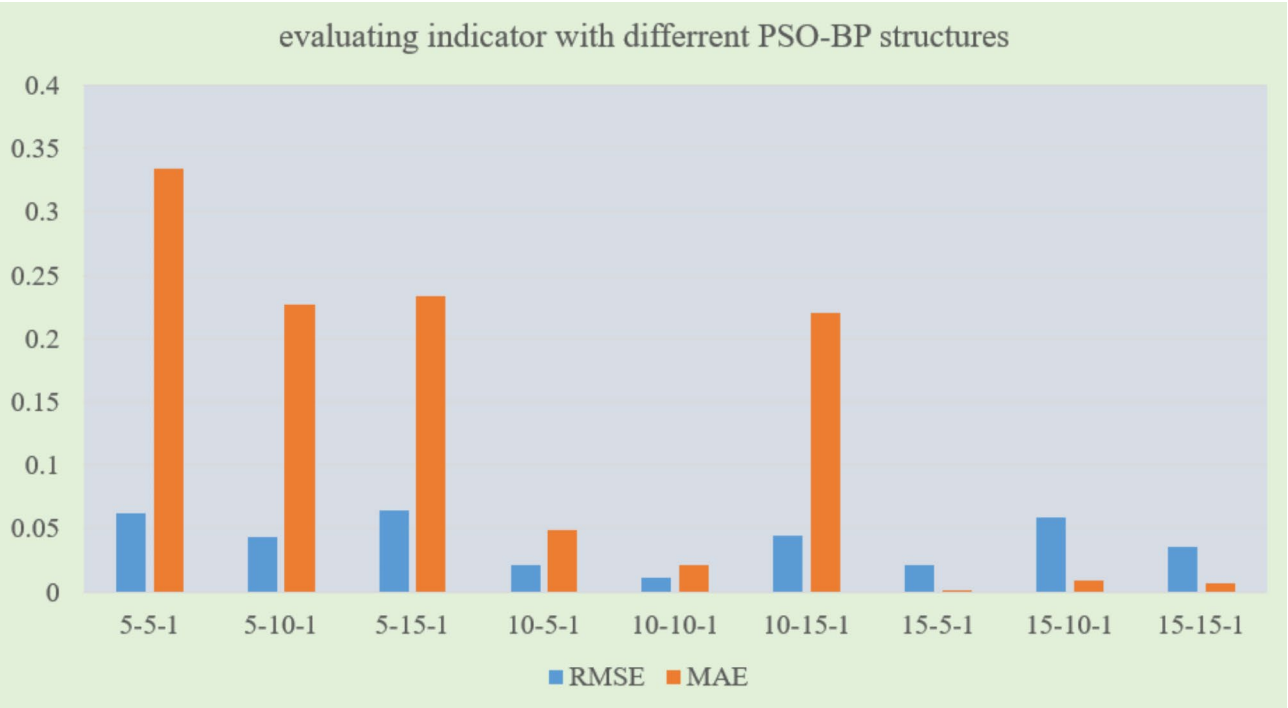


Fig. 11. Comparison of the settlement prediction of DB-3 with different PSO-BP network structures.

0.02%~0.66%, respectively. The LRE for PSO-BP, ARIMA3, ARIMA4, and ARIMA5 are 0.17%, 11.50%, 0.93%, and 1.11%, respectively. It can be seen that the optimal prediction model is the PSO-BP algorithm in most cases. Only in a few instances does the ARIMA3 algorithm slightly outperform the PSO-BP algorithm. This is because, as the rate and magnitude of settlement changes slow down, the fluctuations in the Artificial Neural Network-based method can increase the relative error.

Structures	RS	Structures	RS	Structures	RS
5-5-1	1.9671	10-5-1	0.3269	15-5-1	0.1799
5-10-1	1.2720	10-10-1	0.0602	15-10-1	0.9215
5-15-1	1.6980	10-15-1	1.2718	15-15-1	0.4650

Table 2. Comparison of the RS with different PSO-BP network structures.

	RMSE	MAE
BP + initial data	0.175	0.127
PSO-BP + initial data	0.104	0.087
BP + propcessed data	0.0366	0.0250
PSO-BP + propcessed data	0.0116	0.0213

Table 3. Comparison of the settlement prediction of DB-3 with different datasize.

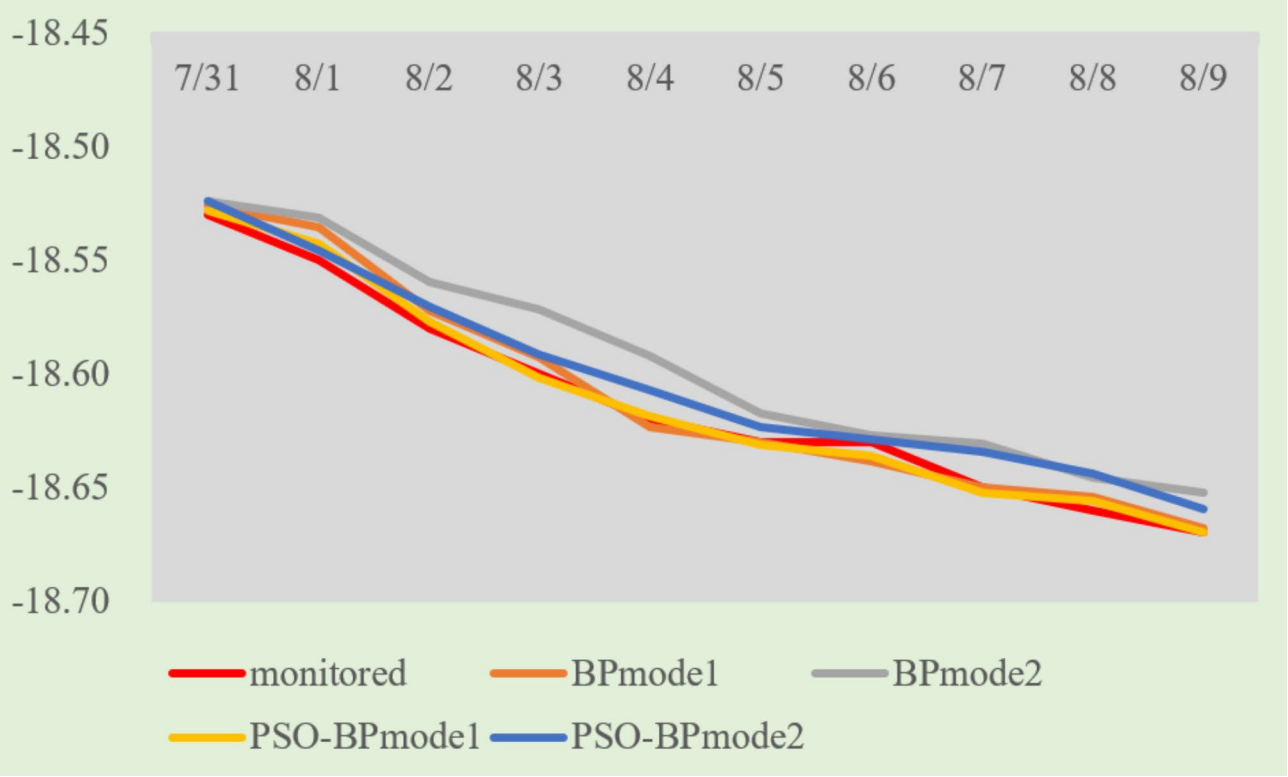


Fig. 12. Comparison of different modes for settlement prediction trend of DB-3.

Conclusion

To predict ground settlement caused by subway shield tunneling construction and provide a reference for ground surface settlement early warning, this study proposes a data preprocessing method based on tangent circles. Utilizing the settlement monitoring data from the LJ to LW section of Wuhan Metro Line 12, prediction analysis was conducted using a BP neural network optimized by the PSO algorithm. The following conclusions can be drawn from this study:

1. Based on the actual monitoring results, the ground surface settlement process can be generally divided into four stages: the uplift stage, the rapid settlement stage, the slow settlement stage, and the stable stage.
2. Through repeated testing of the BP neural network algorithm and the PSO-BP neural network algorithm, it was found that the prediction accuracy is optimal when the input layer, hidden layer, and output layer consist of 10, 10, and 1 node(s), respectively, thus determining the network structure as 10-10-1. Moreover, the PSO-BP neural network algorithm further improves prediction accuracy.

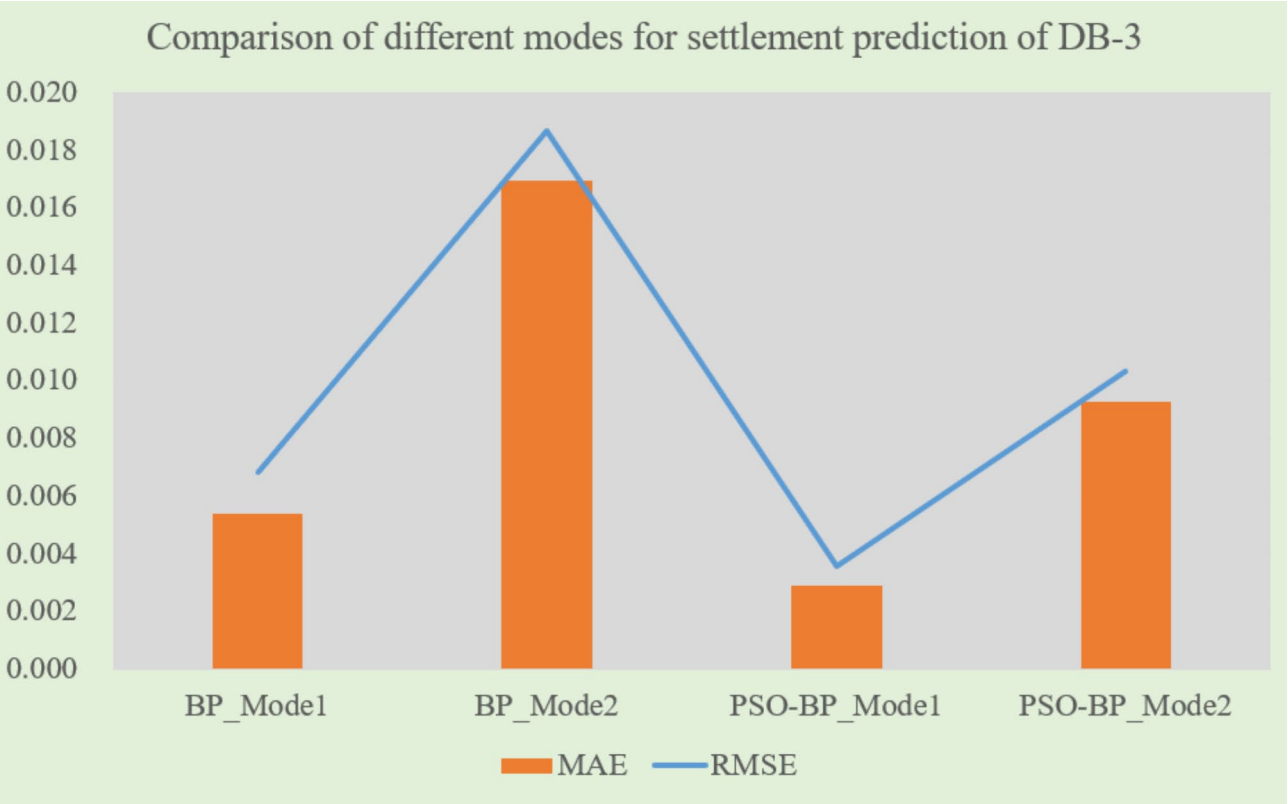


Fig. 13. Comparison of different modes for settlement prediction of DB-3.

3. A comparative analysis of the two prediction modes revealed that predictions based on complete monitoring data (Mode 1) outperform those based on partially new predicted data (Mode 2). However, the error in Mode 2 remains within an acceptable range, and it offers the advantage of long-term prediction, leading to the adoption of Mode 2 as the ground surface settlement prediction mode.
4. The PSO-BP neural network algorithm and the ARIMA algorithm were applied to predict settlement during the slow settlement stage and the stable stage, respectively. The results indicate that the PSO-BP neural network outperforms the ARIMA method in 5-day predictions during the slow settlement stage, with a mean relative error range of 0.13%~0.27%. Compared to the ARIMA method, the PSO-BP neural network algorithm shows higher accuracy during the slow settlement stage. Additionally, during the stable stage, the PSO-BP neural network generally outperforms the ARIMA method in most cases in 5-day predictions; however, in a few instances, the ARIMA3 model slightly outperforms the PSO-BP model in settlement prediction.

In this study, the authors applied a data preprocessing method to achieve the continuous processing of monitoring data and used machine learning algorithms to predict ground settlement induced by metro shield tunneling construction. This can assist the real-time warning and control of ground settlement induced by metro shield tunneling construction. However, the study has the following two limitations: (1) While historical settlement data can represent the trend of soil settlement, the relationship between ground settlement and construction parameters remains unclear, and the correlation between real-time ground settlement data and construction parameters is not well understood; (2) The predicted settlement based on artificial neural network methods contains fluctuations caused by inherent nonlinearity, which reduces prediction accuracy. There are two primary future research directions: (1) Further analysis of the mechanisms underlying ground settlement, exploring the relationship between real-time settlement and construction parameters; (2) Improving the algorithms to enhance the accuracy of long-term ground settlement prediction.

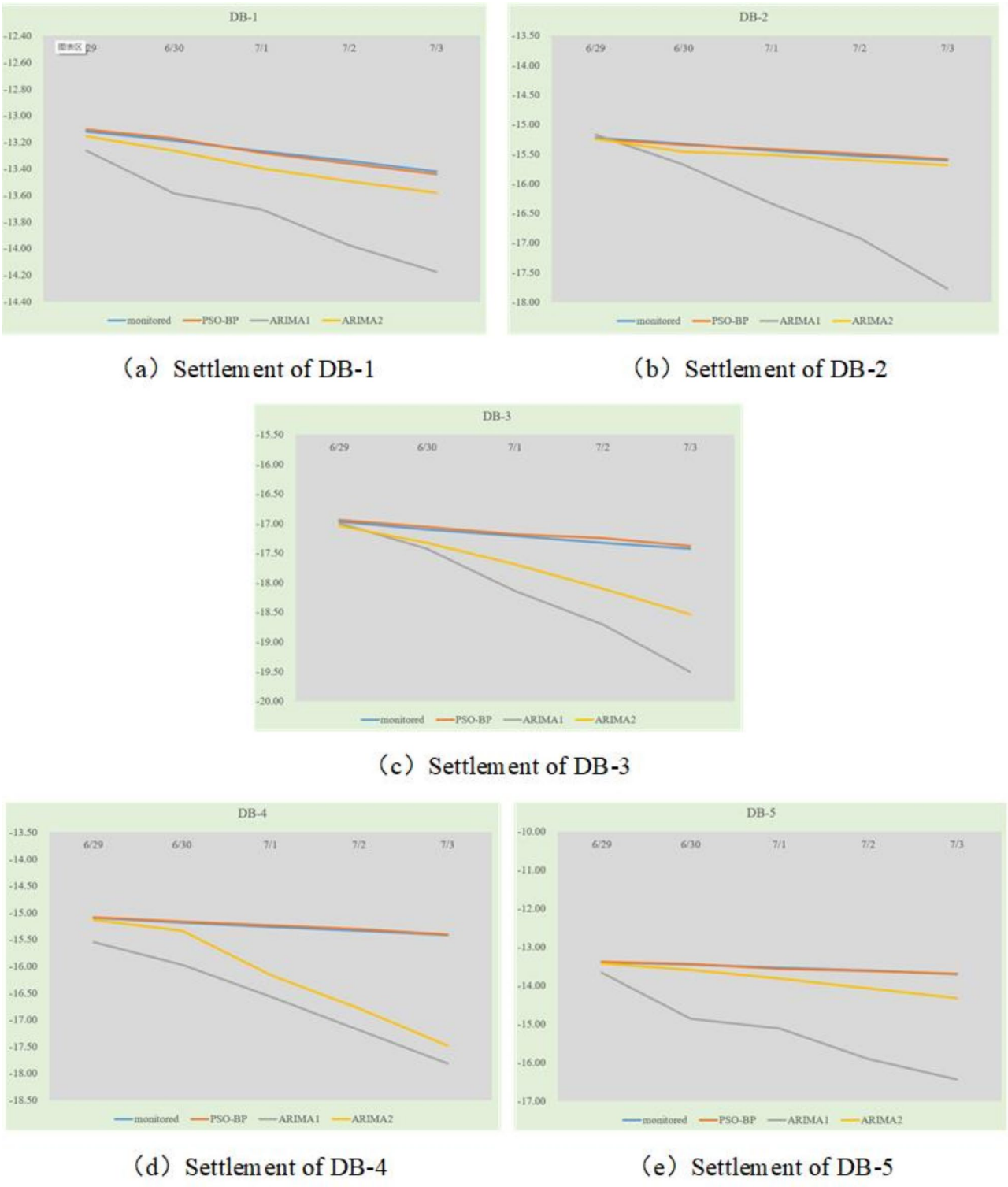


Fig. 14. Forecasting of five-day settlement in the slow settlement stage.

Point	PSO-BP	ARIMA1	ARIMA2
DB-1	0.13%	3.56%	1.63%
DB-2	0.13%	6.25%	0.49%
DB-3	0.27%	5.46%	3.08%
DB-4	0.13%	8.86%	6.00%
DB-5	0.10%	12.21%	2.28%

Table 4. Summary of the MRE for slow settlement forecasting.

Point	PSO-BP	ARIMA1	ARIMA2
DB-1	0.17%	5.63%	2.29%
DB-2	0.18%	13.87%	0.84%
DB-3	0.46%	11.94%	6.36%
DB-4	0.20%	15.53%	13.41%
DB-5	0.16%	20.05%	4.62%

Table 5. Summary of the LRE for slow settlement forecasting.

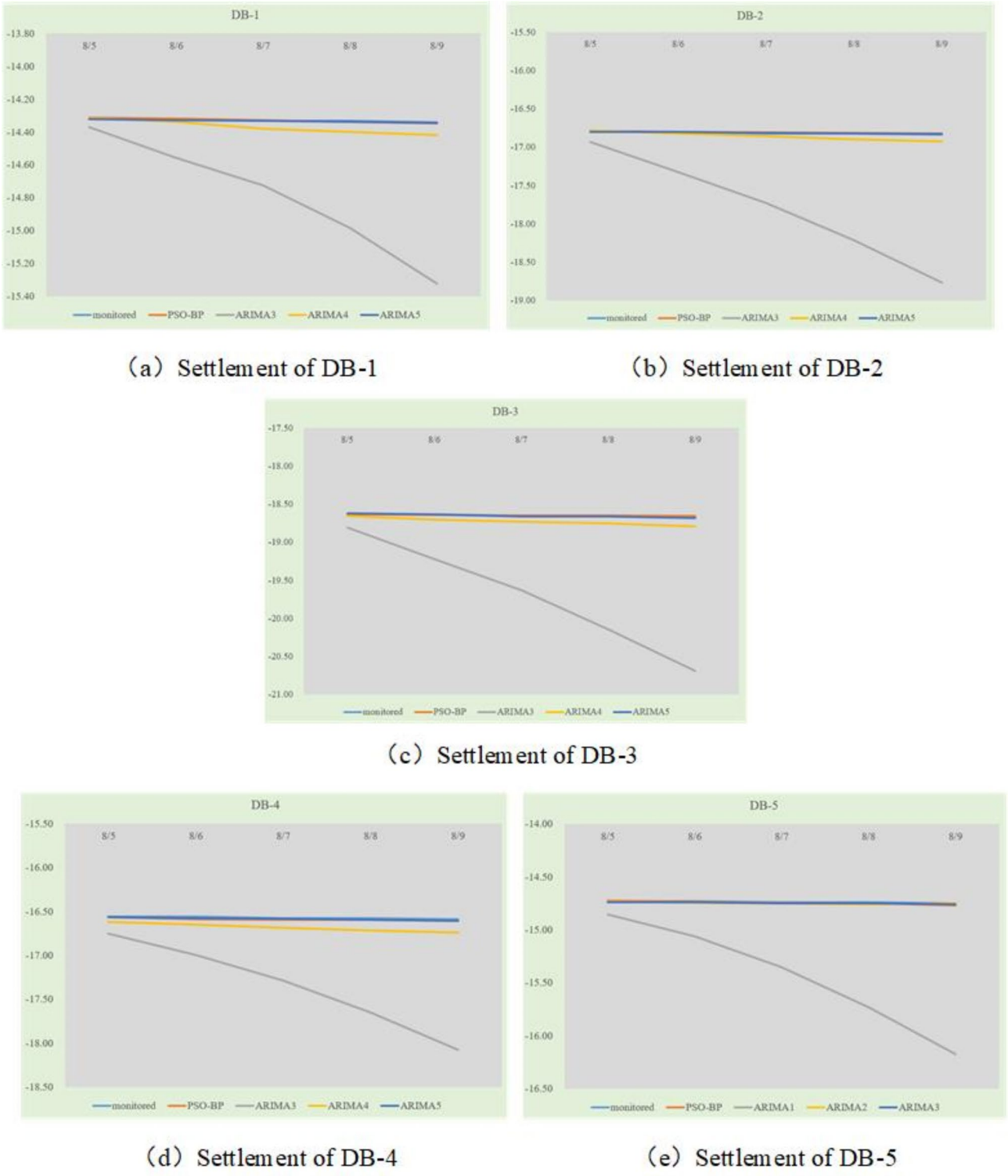


Fig. 15. Forecasting of five-day settlement in the stable stage.

Point	PSO-BP	ARIMA1	ARIMA2	ARIMA3
DB-1	0.04%	3.24%	0.31%	0.04%
DB-2	0.05%	5.82%	0.29%	0.02%
DB-3	0.05%	5.63%	0.41%	0.03%
DB-4	0.10%	4.70%	0.66%	0.66%
DB-5	0.06%	4.73%	0.10%	0.07%

Table 6. Summary of the MRE for stable stage forecasting.

Point	PSO-BP	ARIMA1	ARIMA2	ARIMA3
DB-1	0.06%	6.86%	0.56%	0.06%
DB-2	0.08%	11.50%	0.56%	0.04%
DB-3	0.09%	10.85%	0.65%	0.06%
DB-4	0.17%	8.97%	0.93%	1.11%
DB-5	0.13%	9.65%	0.20%	0.13%

Table 7. Summary of the LRE for stable stage forecasting.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Received: 4 September 2024; Accepted: 9 December 2024
Published online: 30 December 2024

References

1. Qiu, J. L. et al. Structural response of the metro tunnel under local dynamic water environment in loess strata. *Geofluids* https://doi.org/10.1155/2019/8541959 (2019).
2. Wang, F., Ding, L. Y., Luo, H. B. & Love, P. E. D. Probabilistic risk assessment of tunneling-induced damage to existing properties. *Expert Syst. Appl.* **41**, 951–961. https://doi.org/10.1016/j.eswa.2013.06.062 (2014).
3. Wang, W., Gao, S., Min, Y., Liu, L. & Chen, J. Three-dimensional fluid–solid coupling numerical simulation of effects of underlying karst cave on shield tunnel through sand stratum. *Geotech. Geol. Eng.* **37**, 4825–4836 (2019).
4. Wang, Y. X., Wu, T., Li, L. & Huang, Y. D. X. Safety risk assessment of subway shield construction based on catastrophe progression method. *Chin. J. Saf. Environ. Eng.* **28**, 8. https://doi.org/10.13578/j.cnki.issn.1671-1556.2021.01.014 (2021).
5. Meng, F. Y., Chen, R. P. & Kang, X. Effects of tunneling-induced soil disturbance on the post-construction settlement in structured soft soils. *Tunn. Undergr. Space Technol.* **80**, 53–63. https://doi.org/10.1016/j.tust.2018.06.007 (2018).
6. Shi, Y. J., Xiao, X. & Li, M. G. Long-term longitudinal deformation characteristics of metro lines in soft soil area. *J. Aerospace Eng.* **31** https://doi.org/10.1061/(asce)as.1943-5525.0000910 (2018).
7. Su, J., Wang, Y. Z., Niu, X. K., Sha, S. & Yua, J. Y. Prediction of ground surface settlement by shield tunneling using XGBoostand bayesian optimization. *Eng. Appl. Artif. Intell.* **114** https://doi.org/10.1016/j.engappai.2022.105020 (2022).
8. Zhang, Z. G., Huang, M. S. & Zhang, M. X. Theoretical prediction of ground movements induced by tunnelling in multi-layered soils. *Tunn. Undergr. Space Technol.* **26**, 345–355 (2011).
9. Yang, X. L. & Wang, J. M. Ground movement prediction for tunnels using simplified procedure. *Tunn. Undergr. Space Technol.* **26**, 462–471. https://doi.org/10.1016/j.tust.2011.01.002 (2011).
10. Divall, S. & Goodey, R. J. Twin-tunnelling-induced ground movements in clay. *Proc. Institution Civil Engineers-Geotechnical Eng.* **168**, 247–256. https://doi.org/10.1680/geng.14.00054 (2015).
11. Xie, X. Y., Yang, Y. B. & Ji, M. Analysis of ground surface settlement induced by the construction of a large-diameter shield-driven tunnel in Shanghai, China. *Tunn. Undergr. Space Technol.* **51**, 120–132. https://doi.org/10.1016/j.tust.2015.10.008 (2016).
12. Liu, J., Wang, F., He, S. H., Wang, E. Z. & Zhou, H. Enlarging a large-diameter shield tunnel using the Pile-Beam-Arch method to create a metro station. *Tunn. Undergr. Space Technol.* **49**, 130–143. https://doi.org/10.1016/j.tust.2015.04.006 (2015).
13. Shivaei, S., Hataf, N. & Pirastehfar, K. 3D numerical investigation of the coupled interaction behavior between mechanized twin tunnels and groundwater - a case study: Shiraz Metro line 2. *Tunn. Undergr. Space Technol.* **103** https://doi.org/10.1016/j.tust.2020.103458 (2020).
14. Li, Z., Luo, Z. J., Xu, C. H. & Tan, J. Z. 3D fluid-solid full coupling numerical simulation of soil deformation induced by shield tunnelling. *Tunn. Undergr. Space Technol.* **90**, 174–182. https://doi.org/10.1016/j.tust.2019.03.020 (2019).
15. Ritter, S., Giardina, G., DeJong, M. J. & Mair, R. J. Centrifuge modelling of building response to tunnel excavation. *Int. J. Phys. Modelling Geotechnics*. **18**, 146–161. https://doi.org/10.1680/jphmg.16.00053 (2018).
16. Huang, M. Q., Ninic, J. & Zhang, Q. B. BIM, machine learning and computer vision techniques in underground construction: current status and future perspectives. *Tunn. Undergr. Space Technol.* **108** https://doi.org/10.1016/j.tust.2020.103677 (2021).
17. Chen, R. P. et al. Prediction of maximum surface settlement caused by earth pressure balance (EPB) shield tunneling with ANN methods. *Soils Found.* **59**, 284–295. https://doi.org/10.1016/j.sandf.2018.11.005 (2019).
18. Xu, S. T. & Xu, J. M. Prediction of buildings' settlement induced by metro station deep foundation pit construction. *Appl. Sciences-Basel.* **14** https://doi.org/10.3390/app14052143 (2024).
19. Ye, X. W., Jin, T. & Chen, Y. M. Machine learning-based forecasting of soil settlement induced by shield tunneling construction. *Tunn. Undergr. Space Technol.* **124** https://doi.org/10.1016/j.tust.2022.104452 (2022).
20. Zhang, L. M., Wu, X. G., Liu, W. L. & Skibniewski, M. J. Optimal strategy to mitigate tunnel-Induced settlement in soft soils: simulation approach. *J. Perform. Constr. Facil.* **33** https://doi.org/10.1061/(asce)cf.1943-5509.0001322 (2019).
21. Mahmoodzadeh, A. et al. Forecasting maximum surface settlement caused by urban tunneling. *Autom. Constr.* **120** https://doi.org/10.1016/j.autcon.2020.103375 (2020).

22. Zhang, W. G. et al. State-of-the-art review of soft computing applications in underground excavations. *Geosci. Front.* **11**, 1095–1106. https://doi.org/10.1016/j.gsf.2019.12.003 (2020).
23. Zhang, W. G. et al. Soft computing approach for prediction of surface settlement induced by earth pressure balance shield tunneling. *Undergr. Space*. **6**, 353–363. https://doi.org/10.1016/j.undsp.2019.12.003 (2021).
24. Moghtader, T., Sharafati, A., Naderpour, H. & Nik, M. G. Estimating maximum surface settlement caused by EPB shield tunneling utilizing an intelligent approach. *Buildings* **13** https://doi.org/10.3390/buildings13041051 (2023).
25. Ocak, I. & Seker, S. E. Calculation of surface settlements caused by EPBM tunneling using artificial neural network, SVM, and gaussian processes. *Environ. Earth Sci.* **70**, 1263–1276. https://doi.org/10.1007/s12665-012-2214-x (2013).
26. Elbaz, K., Zhou, A. N. & Shen, S. L. Deep reinforcement learning approach to optimize the driving performance of shield tunnelling machines. *Tunn. Undergr. Space Technol.* **136** https://doi.org/10.1016/j.tust.2023.105104 (2023).
27. Bouayad, D. & Emeriault, F. Modeling the relationship between ground surface settlements induced by shield tunneling and the operational and geological parameters based on the hybrid PCA/ANFIS method. *Tunn. Undergr. Space Technol.* **68**, 142–152. https://doi.org/10.1016/j.tust.2017.03.011 (2017).
28. Ge, S. S., Gao, W., Cui, S., Chen, X. & Wang, S. Safety prediction of shield tunnel construction using deep belief network and whale optimization algorithm. *Autom. Constr.* **142** https://doi.org/10.1016/j.autcon.2022.104488 (2022).
29. Zhang, L. M., Wu, X. G., Ji, W. Y. & AbouRizk, S. M. Intelligent approach to estimation of tunnel-induced ground settlement using wavelet packet and support vector machines. *J. Comput. Civil Eng.* **31** https://doi.org/10.1061/(asce)cp.1943-5487.0000621 (2017).
30. Moghaddasi, M. R., Noorian-Bidgoli, M. & ICA-ANN ANN and multiple regression models for prediction of surface settlement caused by tunneling. *Tunn. Undergr. Space Technol.* **79**, 197–209. https://doi.org/10.1016/j.tust.2018.04.016 (2018).
31. Zhang, K., Lyu, H. M., Shen, S. L., Zhou, A. & Yin, Z. Y. Data on evolutionary hybrid neural network approach to predict shield tunneling-induced ground settlements. *Data Brief.* **33**, 106432. https://doi.org/10.1016/j.dib.2020.106432 (2020).
32. Zhang, P., Wu, H. N., Chen, R. P. & Chan, T. H. T. Hybrid meta-heuristic and machine learning algorithms for tunneling-induced settlement prediction: a comparative study. *Tunn. Undergr. Space Technol.* **99** https://doi.org/10.1016/j.tust.2020.103383 (2020).
33. Tang, L. B. & Na, S. Comparison of machine learning methods for ground settlement prediction with different tunneling datasets. *J. Rock Mech. Geotech. Eng.* **13**, 1274–1289. https://doi.org/10.1016/j.jrmge.2021.08.006 (2021).
34. Lin, S. S., Zhang, N., Zhou, A. N. & Shen, S. L. Time-series prediction of shield movement performance during tunneling based on hybrid model. *Tunn. Undergr. Space Technol.* **119** https://doi.org/10.1016/j.tust.2021.104245 (2022).
35. Yazdanparast, M., Koushkgozar, H. A., Hassanpour, J., Kahaki, A. & Khodagholi, M. Predicting maximum settlement induced by EPB shield tunneling through image processing and an intelligent approach. *KSCE J. Civ. Eng.* **28**, 4076–4087. https://doi.org/10.1007/s12205-024-2086-0 (2024).
36. Elbaz, K., Shen, S. L., Zhou, A. N. & Yoo, C. Reinforcement learning-based optimizer to improve the steering of shield tunneling machine. *Acta Geotech.* **19**, 4167–4187. https://doi.org/10.1007/s11440-023-02136-4 (2024).
37. Liu, C. et al. Prediction of surface settlement caused by synchronous grouting during shield tunneling in coarse-grained soils: a combined FEM and machine learning approach. *Undergr. Space*. **16**, 206–223. https://doi.org/10.1016/j.undsp.2023.10.001 (2024).
38. Lu, Y. H., Gong, X. H. & Kipnis, A. B. Prediction of low-energy building energy consumption based on genetic BP algorithm. *Cmc-Computers Mater. Continua*. **72**, 5481–5497. https://doi.org/10.32604/cmc.2022.027089 (2022).
39. Lai, X. P., Tu, Y. H., Yan, B. X., Wu, L. Q. & Liu, X. M. A method for predicting ground pressure in Meihuajing coal mine based on improved BP neural network by immune algorithm-particle swarm optimization. *Processes* **12** https://doi.org/10.3390/pr12010147 (2024).
40. Hu, D. et al. Prediction method of surface settlement of rectangular pipe jacking tunnel based on improved PSO-BP neural network. *Sci. Rep.* **13** https://doi.org/10.1038/s41598-023-32189-0 (2023).

Acknowledgements

This research was supported by 20211h0057C and SKJC-2022-PTDX-021. The authors would like to express their thanks to everyone who assisted with this research.

Author contributions

In this paper，each author's contribution are shown as this：Wang Qiankun: Project administration, Funding acquisition, Resources, Software; Shen Chuxiong: Data curation, Formal Analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing; Tang Chao: Validation, Investigation, Data curation; Guo Zeng: Conceptualization, Methodology, Supervision; Wu Fangqi: Investigation, Visualization, Data curation; Yang Wenyi: Investigation, Data curation.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to C.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2024

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.